Multiplying Bigger Numbers

Three-digit times two-digit: six pieces, or two rows.

243 × 36: six pieces

	200	40	3
30	6,000	1,200	90
6	1,200	240	18

Hundreds, tens and ones across the top; tens and ones down the side.

6,000 + 1,200 + 90 + 1,200 + 240 + 18 = 8,748

The same in two rows

$$\begin{array}{r} 243 \\ \times \quad 36 \\ \hline 1{,}458 \quad (\times 6) \\ 7{,}290 \quad (\times 30) \\ \hline 8{,}748 \end{array}$$

Row 1: 243 × 6 = 1,458 (the bottom pieces).
Row 2: 243 × 30 = 7,290 (the top pieces).
243 × 30 is 243 × 3 tens = 729 tens.
Add the two rows: 8,748.

Your turn!

$$312 \times 23 = \square$$

Multiplying Bigger Numbers: Practice

Split, multiply each piece, add.

Part 1 Hundreds times tens 3 hundreds × 4 tens = 12 thousands.

1) 300 × 40 =	2) 200 × 60 =	3) 500 × 70 =
4) 400 × 30 =	5) 700 × 20 =	6) 600 × 90 =
7) 800 × 50 =	8) 900 × 40 =	9) 200 × 80 =
10) 300 × 70 =	11) 500 × 50 =	12) 400 × 60 =

Multiplying Bigger Numbers: Practice (continued)

Part 2 Six pieces The parts are on the outside. Fill in each piece.

13) $234 \times 15 =$ ☐

	200	30	4
10	☐	☐	☐
5	☐	☐	☐

☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

14) $162 \times 37 =$ ☐

	100	60	2
30	☐	☐	☐
7	☐	☐	☐

☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

15) $425 \times 23 =$ ☐

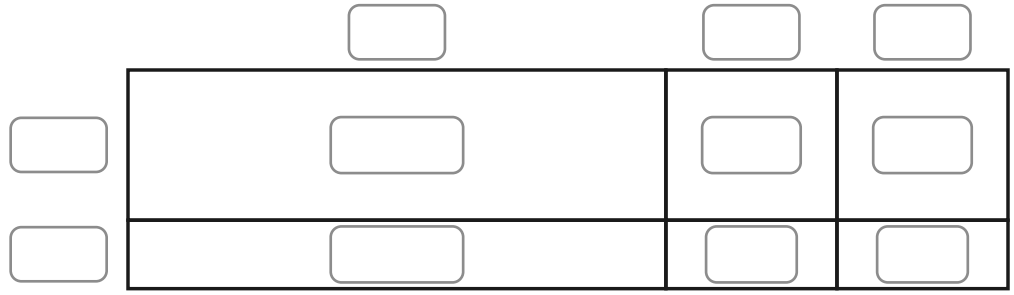
	400	20	5
20	☐	☐	☐
3	☐	☐	☐

☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

Multiplying Bigger Numbers: Practice (continued)

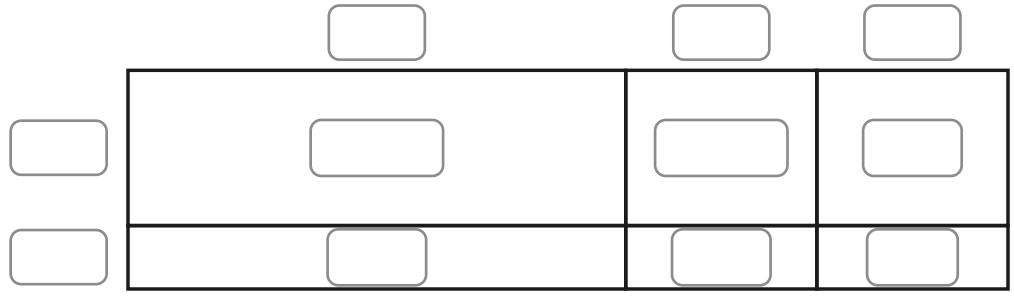
Part 3 Split it yourself Write the parts on the outside, then fill in each piece.

16) $318 \times 24 =$ ☐



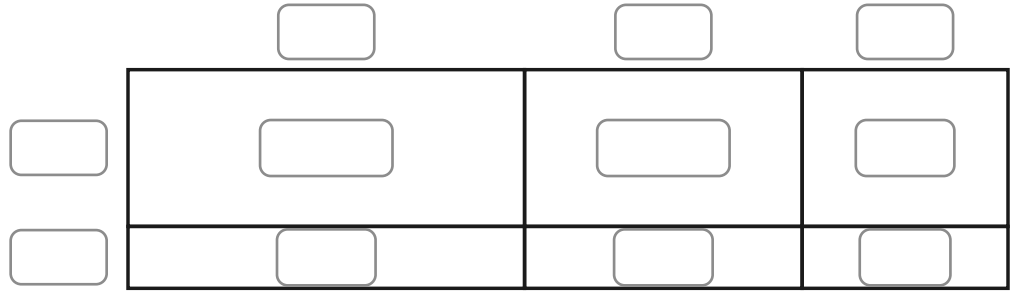
☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

17) $256 \times 43 =$ ☐



☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

18) $173 \times 52 =$ ☐



☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

Multiplying Bigger Numbers: Practice (continued)

Part 4 Two rows Row 1: times the ones. Row 2: times the tens. Then add.

19) 214 × 32

	× 2
	× 30

20) 365 × 24

	× 4
	× 20

21) 128 × 57

	× 7
	× 50

22) 432 × 18

	× 8
	× 10

23) 251 × 46

	× 6
	× 40

24) 617 × 35

	× 5
	× 30

25) 183 × 29

	× 9
	× 20

26) 526 × 14

	× 4
	× 10

27) 342 × 67

	× 7
	× 60

Part 5 Multiply No help this time.

28) 326 × 21 = ____

29) 147 × 38 = ____

Multiplying Bigger Numbers: Practice (continued)

Part 5 Multiply (continued)

30) $512 \times 46 =$ ☐

31) $238 \times 17 =$ ☐

32) $403 \times 25 =$ ☐

33) $671 \times 34 =$ ☐

34) $295 \times 42 =$ ☐

35) $184 \times 63 =$ ☐

36) $736 \times 15 =$ ☐

37) $459 \times 27 =$ ☐

38) $128 \times 74 =$ ☐

39) $387 \times 56 =$ ☐

40) $604 \times 39 =$ ☐

41) $215 \times 88 =$ ☐

Multiplying Bigger Numbers: Practice (continued)

Part 5 Multiply (continued)

42) $548 \times 23 =$ ☐

43) $963 \times 12 =$ ☐

44) $372 \times 45 =$ ☐

45) $819 \times 26 =$ ☐

46) $256 \times 91 =$ ☐

47) $437 \times 68 =$ ☐

Part 6 Four digits times one digit No help this time.

48) $2{,}134 \times 3 =$ ☐

49) $4{,}206 \times 7 =$ ☐

50) $1{,}573 \times 4 =$ ☐

51) $3{,}058 \times 6 =$ ☐

52) $6{,}125 \times 8 =$ ☐

53) $2{,}947 \times 5 =$ ☐

Multiplying Bigger Numbers: Practice (continued)

Part 6 Four digits times one digit (continued)

54) $5{,}312 \times 9 =$ ☐

55) $1{,}806 \times 4 =$ ☐

56) $7{,}243 \times 3 =$ ☐

57) $3{,}615 \times 6 =$ ☐

Part 7 Multiply

58) $672 \times 35 =$ ☐

59) $763 \times 43 =$ ☐

60) $728 \times 41 =$ ☐

61) $163 \times 54 =$ ☐

62) $353 \times 47 =$ ☐

63) $182 \times 39 =$ ☐

64) $672 \times 59 =$ ☐

65) $236 \times 54 =$ ☐

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

66) $526 \times 16 =$ ☐

67) $162 \times 15 =$ ☐

68) $621 \times 38 =$ ☐

69) $113 \times 29 =$ ☐

70) $873 \times 19 =$ ☐

71) $113 \times 22 =$ ☐

72) $194 \times 12 =$ ☐

73) $664 \times 59 =$ ☐

74) $153 \times 33 =$ ☐

75) $542 \times 49 =$ ☐

76) $371 \times 26 =$ ☐

77) $736 \times 48 =$ ☐

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

78) $358 \times 17 =$ ☐

79) $848 \times 59 =$ ☐

80) $237 \times 38 =$ ☐

81) $275 \times 42 =$ ☐

82) $344 \times 58 =$ ☐

83) $355 \times 48 =$ ☐

84) $148 \times 17 =$ ☐

85) $464 \times 12 =$ ☐

86) $497 \times 24 =$ ☐

87) $527 \times 55 =$ ☐

88) $459 \times 14 =$ ☐

89) $604 \times 24 =$ ☐

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

90) $182 \times 36 =$ ☐

91) $883 \times 43 =$ ☐

92) $121 \times 13 =$ ☐

93) $165 \times 47 =$ ☐

Part 8 Remember? Problems from earlier packets.

94) $3.6 + 2.75 =$ ☐

95) $8.2 - 4.67 =$ ☐

96) 4.28 to the nearest tenth = ☐

97) 0.45 ☐ 0.405

98) in 3.142 the 4 is worth ☐

99) $2.7 \times 1{,}000 =$ ☐

Multiplying Bigger Numbers: Practice (continued)

Part 9 Word problems Split, multiply, add.

100) A school bus holds 48 kids. How many kids can 125 buses hold?

Number sentence _____

Answer ☐ kids

Draw or show your work

101) A farm packs 236 boxes of eggs. Each box has 12 eggs. How many eggs is that?

Number sentence _____

Answer ☐ eggs

Draw or show your work

102) A plane flies 465 miles each day for 21 days. How far does it fly?

Number sentence _____

Answer ☐ miles

Draw or show your work

103) A box of paper has 1,250 sheets. How many sheets are in 8 boxes?

Number sentence _____

Answer ☐ sheets

Draw or show your work

104) Challenge: A theater has 34 rows of 28 seats. 875 seats are full. How many are empty?

Number sentence _____

Answer ☐ seats

Draw or show your work

Check-In: Multiplying Bigger Numbers

Show what you know. No help this time. Score: _____ / 10

Part 1 Hundreds times tens

1) $300 \times 90 =$ ☐

2) $800 \times 90 =$ ☐

3) $700 \times 70 =$ ☐

Part 2 Two rows

4)

$$\begin{array}{r} 418 \\ \times \quad 23 \\ \hline \square \quad \times 3 \\ \square \quad \times 20 \\ \hline \square \end{array}$$

5)

$$\begin{array}{r} 152 \\ \times \quad 47 \\ \hline \square \quad \times 7 \\ \square \quad \times 40 \\ \hline \square \end{array}$$

6)

$$\begin{array}{r} 324 \\ \times \quad 26 \\ \hline \square \quad \times 6 \\ \square \quad \times 20 \\ \hline \square \end{array}$$

Part 3 Multiply

7) $382 \times 19 =$ ☐

8) $527 \times 34 =$ ☐

9) $243 \times 17 =$ ☐

10) $275 \times 36 =$ ☐